

Controlling a MuJoCo Dexterous Hand Model Using Spiking Neurons (FEAGI Framework)

Group Name: Group 9

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Introduction

This project demonstrates the integration of MuJoCo with FEAGI to simulate and control the Shadow Hand robotic model in a biologically inspired neural environment.

MuJoCo is a high-performance physics simulator widely used for robotics, reinforcement learning, and AI research. It provides accurate real-time simulation of robotic systems, including articulated hands, humanoids, and manipulators. In this setup, MuJoCo is used to load and visualize the Shadow Hand model inside a simulation environment.

FEAGI is a modular neural execution engine designed for embodied AI and neurorobotics applications. It enables robots and simulators to interact with biologically inspired neural networks through real-time sensory and motor communication. FEAGI also includes a Brain Visualizer that allows users to observe neural activity dynamically during simulation.

The implementation combines:

- MuJoCo physics simulation
- Shadow Hand robotic embodiment
- FEAGI neural control framework
- Brain Visualizer for neural activity monitoring

The overall workflow includes installing MuJoCo, setting up the Shadow Hand model, configuring FEAGI and Brain Visualizer, and connecting the MuJoCo embodiment controller to FEAGI. Once integrated successfully, the simulator can display and control the Shadow Hand using FEAGI-driven neural communication.

This setup is useful for:

- Robotics research
- Embodied AI experiments
- Neural control simulations
- Human hand manipulation studies
- Reinforcement learning environments

Official resources:

- [MuJoCo](#)
- [FEAGI](#)
- [FEAGI GitHub Organization](#)

Official Git Repositories:

- [Mujoco_menagerie](#)
- [FEAGI](#)
- [Embodiment-controllers](#)

MuJoCo installation:-

Prerequisites

- Windows 10/11 (64-bit)
- Python 3.10 or 3.11
- Git installed
- MuJoCo installed via pip

Step 1: Install MuJoCo

Command : pip install mujoco

Step 2 : Verify installation:

Run Python:

python

Then type:

```
import mujoco
print(mujoco.__version__)
```

Note: If no error appears, MuJoCo is installed correctly.

Step 3: Clone the repository:

- Clone the Shadow Hand model:
git clone https://github.com/google-deepmind/mujoco_menagerie
 - The Shadow Hand scene file is at:
mujoco_menagerie/shadow_hand/scene_right.xml
-

Step 4: To view the MuJoCo + Shadow hand in the simulator

```
```python
import mujoco
import mujoco.viewer

model = mujoco.MjModel.from_xml_string("""
<mujoco>
 <worldbody>
 <geom type="plane" size="5 5 0.1"/>
 <body pos="0 0 1">
 <geom type="sphere" size="0.1"/>
 </body>
 </worldbody>
</mujoco>""")
data = mujoco.MjData(model)
with mujoco.viewer.launch_passive(model, data):
 while True:
 mujoco.mj_step(model, data)
```

**Note: Save this code file as Viewer\_test.py**

---

#### Step 5: Once installed, you can load a model of as follows:

```
```python
import mujoco
import mujoco.viewer

model = mujoco.MjModel.from_xml_path(
    "mujoco_menagerie/shadow_hand/scene_left.xml"
)
data = mujoco.MjData(model)
with mujoco.viewer.launch_passive(model, data):
    while True:
        mujoco.mj_step(model, data)
```

Note: save this above code as `run_shadow.py`

Step 6: Get the Shadow Hand Model

The Shadow Hand model is located in:

`mujoco_menagerie/shadow_hand/scene.xml`

This file includes:

- Hand model
 - Meshes
 - Simulation environment
-

Note: Before getting into the `shadow_hand` we need to clone the FEAGI controllers

Installation of FEAGI and FEAGI Brain Visualizers

Step 7: Install FEAGI

git clone <https://github.com/feagi/controllers.git>

Pip install feagi

This Includes:

- FEAGI itself, for running neuronal simulations
 - Brain Visualizer, for real-time 3D neural activity visualization (~196MB)
 - Python bindings for FEAGI libraries (intended for advanced users)
 - FEAGI Agent Python SDK for rapidly making agents for FEAGI
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Step 8: Install feagi-core tis is recommended for production

Pip install feagi-core

This Includes:

- Python bindings for FEAGI libraries (intended for advanced users)
- FEAGI Agent Python SDK for rapidly making agents for FEAGI

Note: Both packages use identical imports (``from feagi import ...``)

What is FEAGI?

FEAGI (Framework for Evolutionary Artificial General Intelligence) is a biologically inspired, modular neural execution engine designed for **embodied AI and robotics**. FEAGI enables spiking-neural-circuit-driven perception, cognition, and control across simulated and physical embodiments, with a strong emphasis on **real-time interaction, modularity, and cross-platform deployment**.

FEAGI serves as the core neural runtime behind **Neurorobotics Studio**, powering a growing ecosystem of reusable neural components ("brains"), tools, and integrations for robotics and physical AI.

The FEAGI Python SDK

The FEAGI Python SDK provides the tools you need to:

- **Connect robots and devices** to FEAGI's neural network
- **Build learning agents** for robots, simulators, and games
- **Visualize neural activity** in real-time with Brain Visualizer
- **Control and manage FEAGI** from Python code
- **Interface with diverse embodiments** through standardized communication protocols

Prerequisites and Virtual Environment (Required)

1. Verify Python

- FEAGI SDK requires python 3.10+.
Run the below command
 - # Windows (PowerShell or CMD)
python -version

2. Create Virtual Environment

- # Windows (PowerShell or CMD)
Python -m venv .venv

3. Activate Virtual Environment

- # Windows PowerShell
.\.venv\Scripts\Activate.ps1
- # Windows CMD
.\.venv\Scripts\activate.bat

4. Upgrade pip (inside active venv)

- python -m pip install --upgrade pip

5. Deactivate When Done

- deactivate

Step 9: Quick Start

Get Start with FEAGI

Pip install “feagi” or python -m pip install feagi

feagi start

feagi bv start

Note: This installs FEAGI with Brain Visualizer, creates default configuration automatically, and launches the visualizer.

Step 10: Configuration Management

Initialize FEAGI environment with default configuration:

feagi init

This creates:

- Configuration: ~/.feagi/config/feagi_configuration.toml
- Genomes directory: ~/Documents/FEAGI/Genomes/ (macOS/Windows)
or ~/FEAGI/genomes/ (Linux)
- Connectomes directory: ~/Documents/FEAGI/Connectomes/ or ~/FEAGI/connectomes/
- Logs and cache directories

Output:

FEAGI environment initialized successfully!

Configuration: ~/.feagi/config/feagi_configuration.toml

Genomes: ~/Documents/FEAGI/Genomes/

Connectomes: ~/Documents/FEAGI/Connectomes/

Logs: ~/.feagi/logs/

Cache: ~/.feagi/cache/

Next steps:

1. Edit configuration if needed:

~/.feagi/config/feagi_configuration.toml

2. Start Brain Visualizer:

feagi bv start

Configuration Options

The feagi init command supports several options:

Generate config only (no directory creation)

```
feagi init --config-only
```

Overwrite existing configuration

```
feagi init --force
```

Generate config in custom location

```
feagi init --output ./my_config.toml
```

Clone the MuJoCo embodiment controller (the official FEAGI-MuJoCo bridge)

Note: Now that FEAGI and the Brain Visualizer are working, it's time to resume implementing the demo.

Quick Start

Step 11: Using an Existing Controller

1. Start FEAGI and Brain Visualizer first (required):

- feagi start
- feagi bv start

2. Clone the repository:

- git clone <https://github.com/feagi/embodiment-controllers.git>
- cd embodiment-controllers

3. Navigate to your controller:

- cd simulators/mujoco

4. Set up Python environment:

- # Windows: venv\Scripts\activate
pip install -r requirements.txt

5. Install the requirements:

- pip install -r requirements.txt

6. Run the controller:

- # Local FEAGI (default 127.0.0.1:8000)
python controller.py
- Remote FEAGI
Python controller.py --ip 192.168.1.100 --port 8000

Or

- python controller.py --ip 192.168.56.1 --port 8000 --feagi-zmq-motor-port 5564 --feagi-zmq-registration-port 30001 --feagi-zmq-sensory-port 5558 --feagi-zmq-connection-timeout-ms 5000 --feagi-zmq-registration-retries 10 --feagi-zmq-heartbeat-interval-s 5.0 --feagi-http-timeout-s 30.0 --model_xml "D:\EashKumar_Ankaiah\Personal_Files\My doc\AIDP\Mujoco\mujoco_menagerie\shadow_hand\right_hand.xml" --agent_id


```
"bXVqb2NvX3NpbXVsYXRpb24AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA" --  
auth-token-b64 "AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA="
```

If everything is working:

- MuJoCo simulator window opens
 - Shadow Hand appears
-

Troubleshooting

1. XML file not found

Make sure path is correct:

mujoco_menagerie/shadow_hand/scene.xml

2. MuJoCo viewer not opening

- Update GPU drivers
 - Install Visual C++ Redistributable
 - Avoid Remote Desktop
-

4. Pip install issues

- Avoid Microsoft Store Python if possible
 - Use virtual environment
-

License

This project follows the licenses of:

- MuJoCo
 - FEAGI
 - Associated repositories
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Results Images:

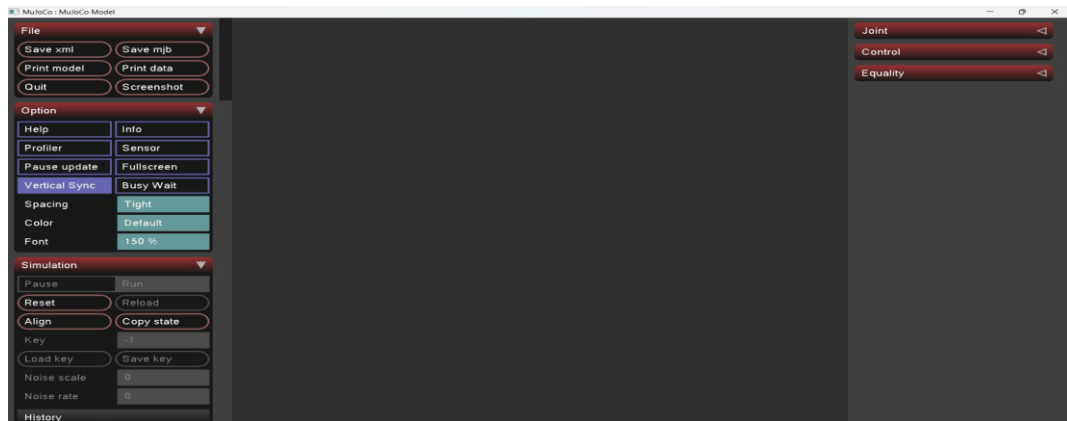
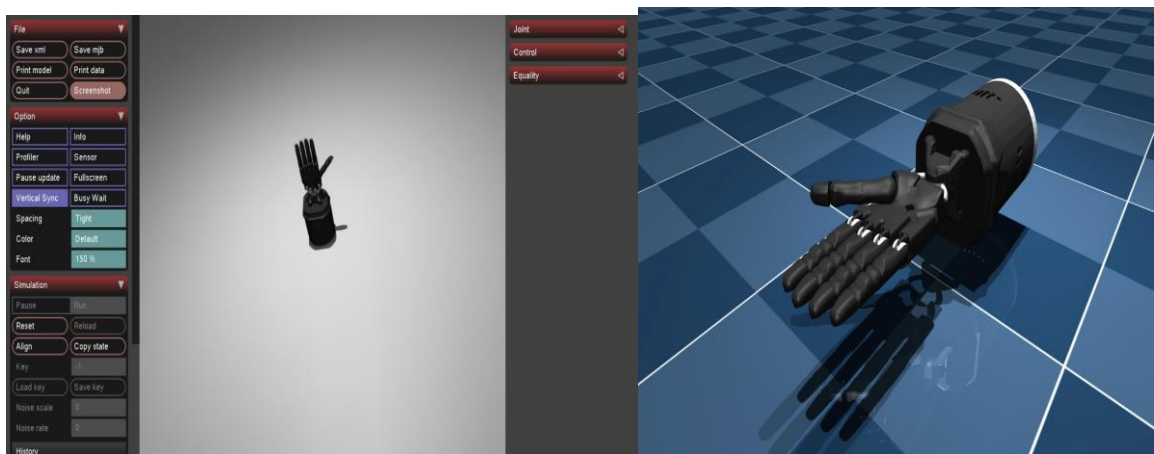


Fig 1 : MuJoCo window



Fig

2: MuJoCo + Shadow hand

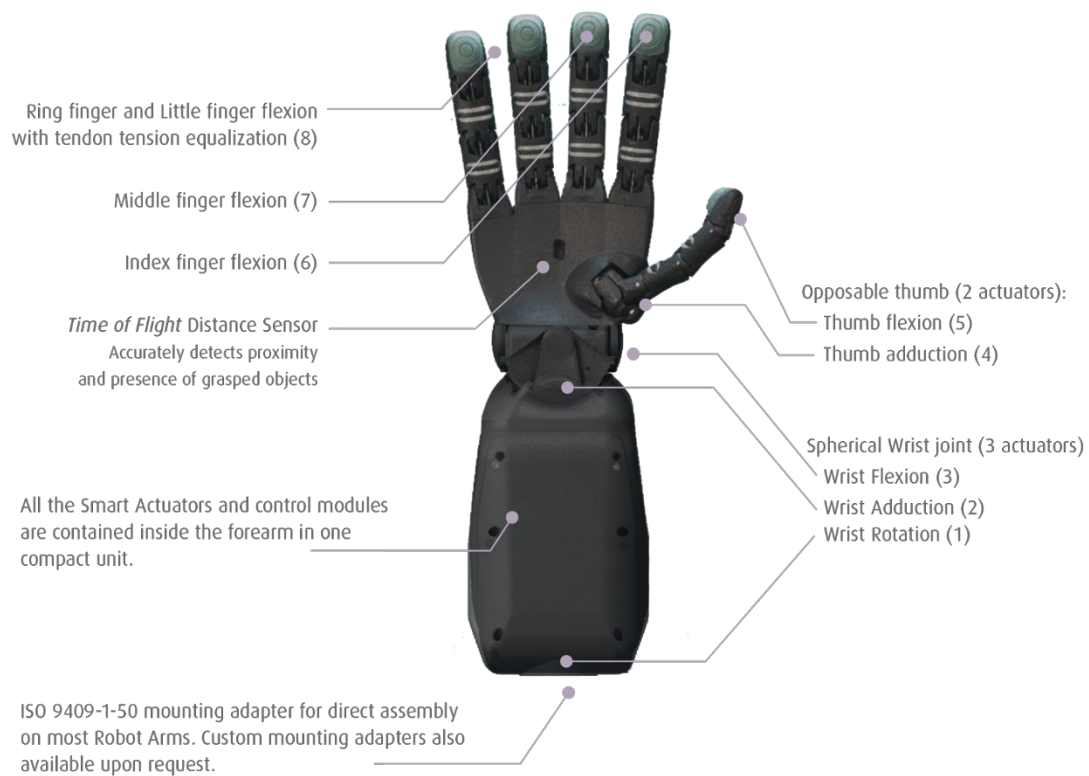
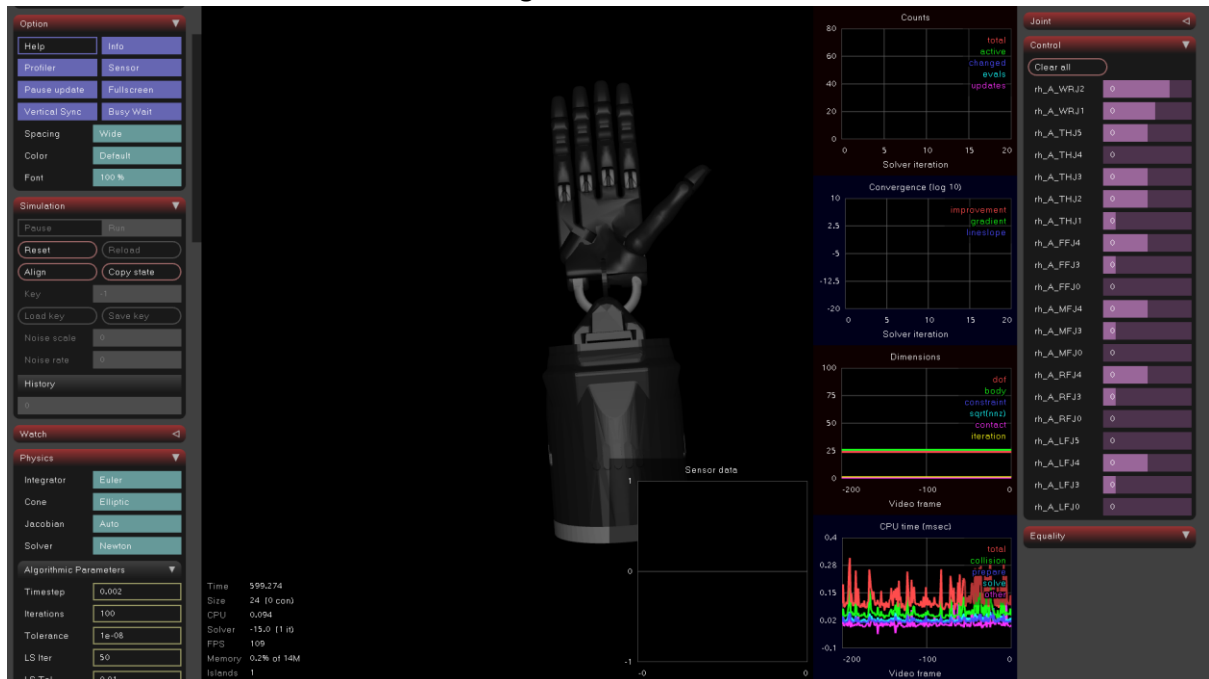


Fig 3: Hand model



```
D:\EashKumar_Ankaiah\Personal_Files\My doc\AIDP\Mujoco\feagi>feagi start
No genome specified, loading barebones genome...
Warning: Could not load barebones genome: HTTPConnectionPool(host='0.0.0.0', port=8000): Max retries exceeded with url:
/v1/genome/upload/barebones (Caused by NewConnectionError('<urllib3.connection.HTTPConnection object at 0x0000020DC1943F
D0>: Failed to establish a new connection: [WinError 10049] The requested address is not valid in its context'))
FEAGI started successfully (PID: 19788)
```

Fig 4: FEAGI successfully running

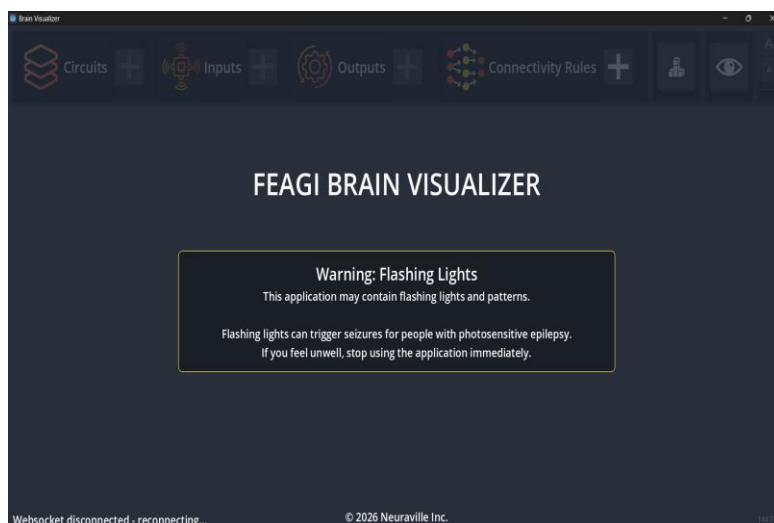


Fig 5: FEAGI Brain Visualizer